

ROBOTIC SURGERY / MEDTECH FIELD OPERATIONS

Bottava Predictive Maintenance

Predict which surgical robots need service — before a failure interrupts a case.

THE CHALLENGE

- Reactive service means unplanned downtime that disrupts surgical schedules.
- Fleet telemetry, case history, and site data live in disconnected silos.
- Teams lack an early warning system to prioritize which robots to service first.

KEY CAPABILITIES

- Component-level maintenance risk score for every robot
- Prioritized watchlist that ranks the fleet by urgency
- Geographic site map for location-aware triage
- Component heatmap isolating stressed parts
- Ask-in-plain-English analytics and AI recommended actions

DATABRICKS PLATFORM

Lakeflow Pipelines

Unity Catalog

MLflow

Model Registry

Mosaic AI

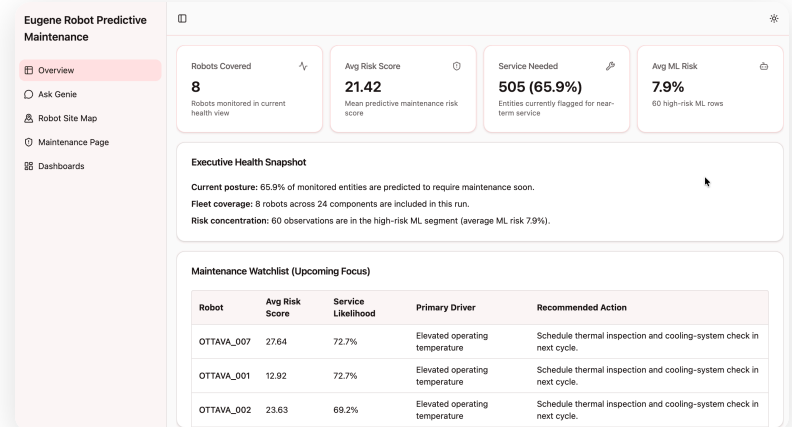
Foundation Model API

Genie

Lakebase

SQL Warehouse

Databricks Apps



AI-POWERED INSIGHT

A machine learning model weighs eight operational signals — errors, thermal stress, vibration, utilization, and service history — to flag at-risk components, then an AI copilot explains each prediction in plain language with recommended next actions.

Predict, don't react

Shift from failure-driven repairs to proactive, scheduled intervention across the fleet.

One governed workflow

Ingestion, ML, dashboards, and an operator app run on a single Unity Catalog-governed platform.

Explainable AI

Every risk score comes with a plain-language rationale and recommended maintenance action.